Business Requirements Document (BRD)

Project Name: UrbanCart

Prepared by: Irfan(TL), Nagraj, Tanvi, Bhubaneswari, Mukunth

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College/University: SMSG Jain Degree College

Guide: Ms. Suchitra Mishra

1. INTRODUCTION .

1.1 Purpose

This document describes the business requirements for an online e-commerce platform that lets customers browse, search, and purchase products, and lets sellers and administrators manage products, orders, and users. It is the basis for the later SRS, design, and development phases.

1.2 Scope

The system is a web application with a customer storefront, a seller panel and an admin panel.

In scope: user registration and login, product catalogue and search, shopping cart, wishlist, checkout, simulated online payment, order tracking, reviews and ratings, seller product management, admin management and reports.

Out of scope (for this version): mobile native apps, real payment gateway with live money, logistics or courier API integration, multi-language and multi-currency support, AI recommendations.

2. Business Overview .

2.1 Background

Online shopping is now a primary way people buy goods. Small sellers often lack an affordable platform to reach customers, and customers want a single place to compare, buy, and track products. This project builds a simplified marketplace modelled on established platforms such as Amazon.

2.2 Problem Statement

Traditional shops are limited by location and opening hours, customers cannot easily compare products and prices, and small sellers lack a low-cost online presence. A web-based platform with product search, secure ordering, and order tracking addresses these gaps.

2.3 Business Objectives

1. Provide a 24x7 online store for browsing and buying products.
2. Allow multiple sellers to list and manage their products.
3. Reduce manual order handling through automated cart, checkout, and order management.
4. Give administrators clear control and visibility over users, products, and orders.
5. Build a secure, scalable base that can be extended later.

2.4 Success Criteria

- A user can register, search, add to cart, check out, and see order status end to end.
- Pages load within 3 seconds under normal load.
- All core modules pass functional testing with no critical open defects.

3. Stakeholders .

Stakeholder	Role	Interest
Customer	End user	Easy browsing, safe payment, fast delivery tracking
Seller	Vendor	List products, manage stock, receive orders
Administrator	System manager	Control users, catalogue, orders, reports
Project Guide	Evaluator	Quality of analysis, design, and implementation
Developer (Student)	Builder	Deliver a working, documented system

4. Current System and Proposed System .

4.1 Existing System

Purchases happen mainly in physical shops or through phone and messaging orders. Limitations: limited reach, manual record keeping, no automatic stock updates, no order history, and no easy price comparison.

4.2 Proposed System

A web platform where customers search and buy online, sellers manage their own listings, and admins oversee everything. Orders, stock, and payments are tracked automatically in a central database.

4.3 Advantages

- Available anytime, from anywhere
- Automatic stock and order tracking
- Faster, more accurate order processing
- Searchable, filterable catalogue
- Centralised reporting

4.4 Justification for the Project

Large e-commerce platforms already exist, so this project is justified by its purpose and focus, not by trying to replace them.

1. **Academic and skill value:** E-commerce brings together authentication, role-based access, database design, transactions, payments, and security. Building it demonstrates the full software development lifecycle on a real-world problem.
2. **Support for small and local sellers:** The platform gives small sellers a simple, low-cost way to list and sell products without the fees, competition, and complexity of the large marketplaces.
3. **Simpler and focused experience:** A smaller catalogue and a cleaner workflow make browsing, ordering, and seller management easier than on a general-purpose giant.
4. **Full control and customisation:** Because the platform is built in-house, features such as local-language support, local delivery rules, and custom reports can be added without depending on a third party.
5. **Proven demand, lower risk:** Online shopping is a proven model with clear user needs, so the project can follow established practices while keeping its scope manageable for a student team.
6. **Extensible foundation:** The modular design allows later additions such as a real payment gateway, recommendations, and a mobile app.

5. User Roles, Permissions and Requirements .

5.1 Role and Permissions:

Role	Key Permissions

Guest	Browse products, search, view details
Customer	All guest rights plus cart, wishlist, checkout, orders, reviews, profile
Seller	Add, edit, delete own products; manage stock; view and process own orders
Admin	Manage all users, sellers, categories, products, orders; approve sellers; view reports

5.2 User (Customer) Requirements:

These are the needs a customer must be able to meet on the platform; Section 6 turns them into detailed functional requirements.

ID	Requirement	Priority
UR-1	Register and log in securely with email or phone and a password	Must
UR-2	Reset a forgotten password through email verification	Must
UR-3	Browse products by category and sub-category	Must
UR-4	Search products by keyword and filter by price, category, and rating	Must
UR-5	View full product details: images, price, description, stock status, reviews	Must
UR-6	Add items to the cart, change quantities, and remove items	Must
UR-7	Manage profile details and multiple delivery addresses	Must
UR-8	Place an order and pay by Cash on Delivery or simulated online payment	Must
UR-9	Receive an order confirmation and invoice	Must
UR-10	View order history and track the current status of each order	Must
UR-11	Use the site comfortably on mobile, tablet, and desktop	Must
UR-12	Save products to a wishlist for later	Should

UR-13	Cancel an order before it is shipped	Should
UR-14	Rate and review products they have purchased	Should
UR-15	Receive email notifications about orders and account activity	Could

Priority key: Must = needed for the first working version; Should = important, include if time allows; Could = optional extra.

5.2 Admin Requirements:

The administrator needs full control over users, catalogue, and orders, plus the information to run the platform.

ID	Requirement	Priority
AR-1	Log in through a secure admin account that is separate from customer accounts	Must
AR-2	See a dashboard with totals for users, orders, and sales	Must
AR-3	View, block, unblock, and delete user accounts	Must
AR-4	Add, edit, and delete categories and sub-categories	Must
AR-5	View, edit, and remove any product listing	Must
AR-6	View all orders and update their status	Must
AR-7	Approve or reject seller registrations	Should
AR-8	Monitor stock levels and get low-stock alerts	Should
AR-9	Moderate reviews and remove inappropriate ones	Should
AR-10	Generate reports: sales, top-selling products, user growth	Should
AR-11	Manage payment and refund records (simulated payments)	Could
AR-12	Keep an audit log of important admin actions	Could

6. Functional Requirements .

6.1 User Management

- FR-1: Users shall register with name, email, phone, and password.
- FR-2: Users shall log in and log out securely.
- FR-3: Users shall reset a forgotten password via email verification.
- FR-4: Users shall view and edit their profile and manage multiple delivery addresses.

6.2 Product Catalogue and Search

- FR-5: The system shall display products organised by category and sub-category.
- FR-6: Each product page shall show images, title, price, description, stock status, and reviews.
- FR-7: Users shall search by keyword and filter by category, price range, and rating.
- FR-8: Users shall sort results by price, popularity, or newest.

6.3 Shopping Cart and Wishlist

- FR-9: Customers shall add, remove, and change quantity of items in the cart.
- FR-10: The cart shall show item totals, taxes, and the grand total.
- FR-11: Customers shall save products to a wishlist and move them to the cart.

6.4 Checkout and Payment

- FR-12: Customers shall select or add a delivery address at checkout.
- FR-13: The system shall support Cash on Delivery and a simulated online payment (test mode).
- FR-14: The system shall validate stock before confirming an order.
- FR-15: The system shall generate an order confirmation and an invoice.

6.5 Order Management

- FR-16: Customers shall view order history and current status (Placed, Confirmed, Shipped, Delivered, Cancelled).
- FR-17: Customers shall cancel an order before it is shipped.
- FR-18: Sellers shall update the status of orders for their products.

6.6 Reviews and Ratings

- FR-19: Customers who purchased a product shall rate and review it.
- FR-20: The product page shall show the average rating and individual reviews.

6.7 Seller Module

- FR-21: Sellers shall register and await admin approval.

- FR-22: Approved sellers shall add, edit, and remove products with images and stock quantity.
- FR-23: Sellers shall view sales and order summaries.

6.8 Admin Module

- FR-24: Admins shall manage users, sellers, categories, and products.
- FR-25: Admins shall approve or reject seller applications.
- FR-26: Admins shall view and manage all orders.
- FR-27: Admins shall view reports: sales, top products, user growth.

6.9 Notifications

- FR-28: The system shall send email notifications for registration, order confirmation, and status changes.

7. Non-Functional Requirements .

Category	Requirement
Performance	Pages load in about 3 seconds or less under normal load; search results in under 2 seconds
Security	Passwords hashed (bcrypt), token-based authentication (JWT), HTTPS, protection against SQL injection, XSS, and CSRF, rolebased access control
Usability	Clean, intuitive interface; consistent navigation
Responsiveness	Works on desktop, tablet, and mobile browsers
Reliability	Database transactions keep orders and stock consistent; regular backups
Scalability	Modular design so features can be added without rewriting the system
Maintainability	Clean code structure, API documentation, version control (Git)

Category	Requirement
Compatibility	Latest versions of Chrome, Firefox, Edge, and Safari

8. Business Rules

- BR-1: A product cannot be ordered if stock is zero.
- BR-2: An order can be cancelled only before it is shipped.
- BR-3: Only customers who bought a product may review it.
- BR-4: Sellers can modify only their own products and orders.
- BR-5: Seller accounts become active only after admin approval.
- BR-6: Email addresses must be unique per account.

9. Assumptions and Constraints

9.1 Assumptions

- Users have internet access and a modern web browser.
- Payment is simulated; no real money is processed.
- Product data is entered by sellers or loaded as sample data.

9.2 Constraints

- The project must be completed within the academic schedule.
- Only the approved technologies may be used.
- Developed by a single student (or small team) with limited budget.

10. Proposed Technology Stack

Layer	Technology
Frontend	HTML5, CSS3, JavaScript, React
Backend	Python, FastAPI (REST APIs)
Database	MySQL
Authentication	JWT, bcrypt
Tools	Git and GitHub, VS Code, Postman

11. High-Level Architecture

The React frontend communicates with the FastAPI backend over REST APIs (JSON). The backend applies business logic and reads and writes data in the MySQL database.
Authentication tokens are issued by the backend and verified on each protected request.

Flow: Browser (React) -> REST API (FastAPI) -> MySQL Database

12. Major Modules .

1. Authentication and User Management
2. Product and Category Management
3. Search and Filter
4. Cart and Wishlist
5. Checkout and Payment
6. Order Management
7. Reviews and Ratings
8. Seller Panel
9. Admin Panel and Reports
10. Notifications

13. Project Timeline (Proposed) .

Phase	Duration
Requirement analysis and BRD	Weeks 1-2
SRS and system design (ER diagram, DFD, UML)	Weeks 3-4
Database and backend development	Weeks 5-8
Frontend development	Weeks 9-12
Integration and testing	Weeks 13-14
Documentation and final submission	Weeks 15-16

15. Future Enhancements .

- Real payment gateway integration (Razorpay or Stripe)
- Mobile application
- Coupons and discounts
- Delivery partner integration and live tracking